```
def print_board(b):
    for i in range(0, 9, 3):
        print(b[i], '|', b[i+1], '|', b[i+2])
    print()
def check_winner(b):
    wins = [(0,1,2),(3,4,5),(6,7,8),
            (0,3,6),(1,4,7),(2,5,8),
            (0,4,8),(2,4,6)]
    for a, b_, c in wins:
        if b[a] == b[b_] == b[c] != ' ':
            return b[a]
    return 'Draw' if ' ' not in b else None
# Minimax with Alpha-Beta Pruning
def minimax(b, max_player, player, alpha, beta):
    opp = 'O' if player == 'X' else 'X'
    w = check_winner(b)
    if w == player: return 1
    if w == opp: return -1
    if w == 'Draw': return 0
    if max_player:
        best = -float('inf')
        for i in range(9):
            if b[i] == ' ':
                b[i] = player
                val = minimax(b, False, player, alpha, beta)
                b[i] = ' '
                best = max(best, val)
                alpha = max(alpha, best)
                if beta <= alpha:
                    break  # Beta cut-off
        return best
    else:
        best = float('inf')
        for i in range(9):
            if b[i] == ' ':
                b[i] = opp
                val = minimax(b, True, player, alpha, beta)
                b[i] = ' '
                best = min(best, val)
                beta = min(beta, best)
                if beta <= alpha:
                    break  # Alpha cut-off
        return best
def best_move(b, player):
    move, best_score = None, -float('inf')
    for i in range(9):
        if b[i] == ' ':
            b[i] = player
            score = minimax(b, False, player, -float('inf'), float('inf'))
            b[i] = ' '
            if score > best_score:
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            best_score, move = score, i
    return move
def play(player):
    board = [' '] * 9
    turn = 'X'
    print("Game start:")
    print_board(board)
    while True:
        if turn == player:
            m = best_move(board, player)
            print(f"Player ({player}) moves to {m}")
        else:
            m = best_move(board, 'O' if player == 'X' else 'X')
            print(f"Opponent ({turn}) moves to {m}")
        board[m] = turn
        print_board(board)
        w = check_winner(board)
        if w:
            if w == 'Draw': print("Game ended in a Draw.")
            elif w == player: print(f"Player ({player}) wins!")
            else: print(f"Opponent ({turn}) wins!")
            break
        turn = 'O' if turn == 'X' else 'X'
# Run the game
play('O')  # Change to 'X' if you want the AI to play as Xge to 'X' to make AI play X
```